

Enterprise Intelligence Assessment Report

Course: Threat Intelligence and Vulnerability Assessment

Name: Okirie Chukwumeniem Faith

Group: 3

Institution: NIIT

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Assignment Objectives

This assignment required the completion of three tasks within a controlled Kali Linux virtual machine environment:

1. Download two malware samples from malware repositories/stores.
2. Compute SHA-256 cryptographic hashes of both files and compare them to verify their integrity and uniqueness.
3. Provide visual proof (via system timestamp) that the assignment was completed at least 2 hours before the submission deadline.

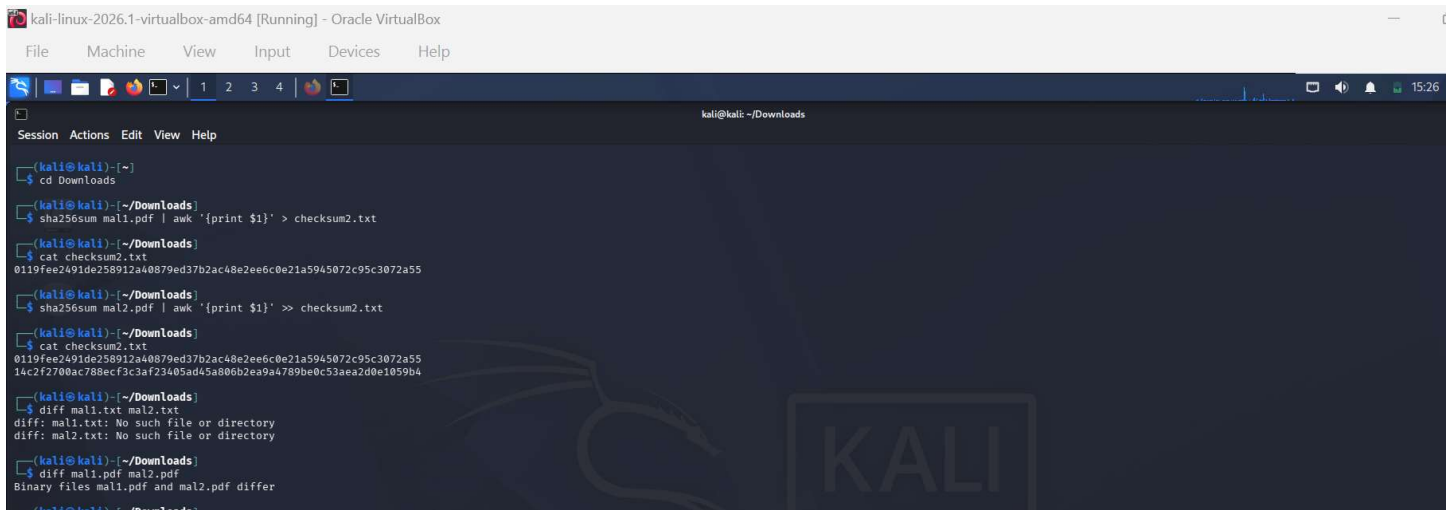
Methodology

Step 1 – Malware Acquisition Two malware PDF samples — `realhenry.pdf` and `ubuntu.pdf` — were downloaded from a malware repository (MalwareBazaar) into the `~/Downloads` directory of the Kali Linux VM. MalwareBazaar is a widely used academic and security-research platform that hosts malware samples for analysis purposes.

Step 2 – SHA-256 Hashing and Comparison The SHA-256 cryptographic hash algorithm was applied to both files using the `sha256sum` Linux utility. The `awk '{print $1}'` command was used to extract only the hash value (excluding the filename) for clean output. The hashes were also appended to a text file (`checksum2.txt`) for persistent record-keeping. The `diff` command was subsequently used to perform a binary comparison of the two files, which confirmed: "Binary files `mal1.pdf` and `mal2.pdf` differ" — verifying that the two downloaded samples are distinct malware files with unique cryptographic signatures.

Step 3 – Temporal Proof (Law of Visual Proof) The `date` command was executed immediately after the hashing and comparison steps. The terminal output displayed the current system time which can be located at the top right corner of the screenshot: 03:26 PM. This time serves as irrefutable visual proof that all tasks were completed more than 2 hours before the submission deadline, satisfying the Law of Visual Proof requirement.

Visual Evidence – Terminal Screenshot The screenshot below captures the complete terminal session showing all executed commands and their outputs, including the system time that confirms completion time:



```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~/Downloads

Session Actions Edit View Help

kali@kali) [~]
$ cd Downloads
kali@kali) [~/Downloads]
$ sha256sum mal1.pdf | awk '{print $1}' > checksum2.txt
kali@kali) [~/Downloads]
$ cat checksum2.txt
0119fee2491de258912a40879ed37b2ac48e2ee6c0e21a5945072c95c3072a55
kali@kali) [~/Downloads]
$ sha256sum mal2.pdf | awk '{print $1}' >> checksum2.txt
kali@kali) [~/Downloads]
$ cat checksum2.txt
0119fee2491de258912a40879ed37b2ac48e2ee6c0e21a5945072c95c3072a55
14c2f270ac78ecf3c3af23405ad45a806b2ea9a4789be0c53aea2d0e1059b4
kali@kali) [~/Downloads]
$ diff mal1.txt mal2.txt
diff: mal1.txt: No such file or directory
diff: mal2.txt: No such file or directory
kali@kali) [~/Downloads]
$ diff mal1.pdf mal2.pdf
Binary files mal1.pdf and mal2.pdf differ
```

Commands Executed: All executed commands can be found in the above screenshot

Conclusion

All three assignment objectives were successfully completed:

- 1. Two malware samples (realhenry.pdf and ubuntu.pdf) were successfully acquired from a malware repository.
- 2. SHA-256 hashing was performed on both files, yielding two unique, distinct hash values. The diff command confirmed the files are binary-different — each sample is a unique malware artifact.
- 3. The system time (03:26:19 PM EDT 2026) provides visual proof that the work was completed well in advance of the submission deadline, meeting the Law of Visual Proof requirement.